VS Code Copilot network-egress redirect — components and calls

kernel mode
user mode (everything else)

KERNEL MODE — zsai-netflt.sys (loaded driver)

ConnectRedirectClassify
ALE_CONNECT_REDIRECT_V4 / _V6
redirect only — never blocks
allocates ZSAI_NETFLT_ORIGINAL_DEST

StreamClassify + FlowEstablished
STREAM / ALE_FLOW_ESTABLISHED
SNI capture, pend-to-user-mode
separate feature, same driver

USER MODE

VS Code Copilot process
copilot-chat ext host +
@github/copilot agent host

ai-gateway listener
kinds::https.rs + certs.rs
mints leaf, terminates TLS,
relays through Optimus

Optimus
policy enforcement backend

real destination (external)
api.githubcopilot.com / api.github.com

ai-protect daemon
net_egress: reconcile.rs, callout.rs
agent-manager: CA provisioning
(.broker.zip)

4 — 5 — 6 — 7 — 8 — 9 — 3 — 1 — 2 — 10

Legend

① ai-protect daemon adds the redirect filter: FwpmFilterAdd0, filter->context = pid<<16 | port — SETUP, before any traffic

② ai-protect daemon delivers the CA (cert_pem + key_pem) to the listener via .broker.zip — SETUP

③ ai-protect daemon installs the root CA into the agent's trust (NODE_EXTRA_CA_CERTS or OS store)

④ connect() to api.githubcopilot.com:443 — client-initiated, unmodified at this point

⑤ ConnectRedirectClassify rewrites dest to 127.0.0.1:<port>, stamps localRedirectTargetPID / Handle / Context

⑥ Listener recovers the original dest via SIO_QUERY_WFP_CONNECTION_REDIRECT_CONTEXT — not yet implemented

⑦ TLS handshake: listener mints a per-host leaf (certs.rs); client validates it against the root CA installed in ③

⑧ Listener relays decrypted, policy-visible traffic onward

⑨ Optimus forwards the request on, post-policy-decision

⑩ Separate (SNI) feature, same driver: daemon ↔ StreamClassify via IOCTL (independent of ①–⑨)

①–③ are one-time setup, done before any connection attempt; ④–⑨ repeat per connection. ①'s filter-add is real code but idle today (finding #1, held); ②'s kernel code (finding #9) is uncompiled/untested; ⑥ has no caller yet.

⑩'s SNI enforcement path is an independent, already-shipped feature sharing this driver, running continuously rather than as a step in ①–⑨'s sequence.